Pre-Registration

Poisson-Certified Muon Counting: A Hardware-Validated, Dead-Time-Corrected Channel Applied to the Positive-Muon Lifetime — Protocol v1.1

Nasir Khan
Wana Lab, Wana, South Waziristan, Pakistan
ORCID: 0009-0002-6006-0796 ·
nasirk684@gmail.com

Protocol version: 1.1 (amendment of v1.0)
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Status: Locked. Amendments require a versioned addendum with a new time-stamp and restart of run numbering.

Changelog v1.0 → v1.1 *(deposited before any physics data; no dataset existed at the time of this amendment, so analysis integrity is unaffected)*:

1. Study aim re-scoped: the project is a **methods and validation study** of a Poisson-certified counting channel; the muon lifetime is the validation observable, compared against the PDG value — not advanced as a competitive precision measurement.
2. New Sec. 3: five pre-specified calibration datasets (CD-1 … CD-5) defining the instrument-chain calibration protocols and acceptance criteria.
3. Expected-outcome section re-scoped

accordingly.

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1. Study aim

Demonstrate, on a real physics signal, an end-to-end **Poisson-certified muon counting chain**: event counting, anti-coincidence veto, and rate calibration performed by the hardware-verified, dead-time-corrected channel QCID-Sigma v5.4 [2], with the Poisson character of the resulting decay-time spectrum certified by the Fano and parametric-bootstrap Kolmogorov–Smirnov protocol (Sec. 6), and a pre-registered unblinding gate (Sec. 7). The validation observable is the positive-muon lifetime τ^{+}_{μ}, compared to the PDG value [1]. No claim of literature-competitive

precision on $\tau^+\mu$ is made or implied; the projected precision (Sec. 9) is presented as a projection only.

2. Apparatus (fixed)

- Two scintillator trigger paddles and one plastic scintillator stopper (5 × 5 × 5 cm^3), PMT/SiPM readout.
- Fast TDC (1 ns bins, FPGA-based), calibrated against a GPS-disciplined oscillator; used exclusively for decay-time stamping.
- QCID-Sigma v5.4 channel on singles monitors: counting, veto, and dead-time-corrected rate calibration only. Not used for decay-time stamping (verified timing envelope median 12 μs is insufficient for the 2.2 μs decay scale).

- Reference instruments for cross-checks: commercial scaler and commercial TDC (models recorded in the run log).
- Expected stopped-muon rate: 5–15 $\mathrm{min^{-1}}$ (sea-level vertical flux ≈ 1 $\mathrm{cm^{-2}\ min^{-1}}$).

3. Calibration datasets (pre-specified)

Each dataset has a number, a protocol, and an acceptance criterion, and is logged and archived identically to physics data. Failure of an acceptance criterion voids the affected runs and triggers a versioned amendment before replacement data are taken.

- **CD-1 — TDC calibration.** Delay-line sweep spanning 0–20 μs in 0.5 μs steps, 10^4 pulses per point, before first physics data and

weekly thereafter; daily GPSDO frequency audit. *Acceptance:* integral nonlinearity < 0.1% of full scale; day-to-day drift < 50 ppm.

- CD-2 — Threshold/energy response. Minimum-ionizing cosmic-muon spectra at discriminator thresholds −20%, nominal, +20% about the operating point, ≥ 10^4 events each. *Acceptance:* stopped-event selection efficiency varies < 0.5% (absolute) across the scan; otherwise the operating point is re-fixed and the scan repeated.
- CD-3 — Afterpulse/late-pulse characterization. Pulsed-LED dataset, 10^5 pulses at the nominal threshold, post-pulse time spectrum recorded to 20 μs. This dataset alone governs the late-pulse decision rule of Sec. 5. *Acceptance:* CD-3 must exist and be cited in the fit report; no post-hoc late-pulse decisions.
- CD-4 — Dead-time model audit. Pulsed-

source rate scan at 1.2, 10, and 50 min^{-1} each run week, against the non-paralyzable model $\lambda_{obs} = \lambda/(1 + \lambda\tau_{dt})$ with $\tau_{dt} = 0.050$ s fixed to the verified value of [2]. *Acceptance:* recovered correction at 10 min^{-1} within 0.83% ± 0.1%.

- CD-5 — Campaign stability. Continuous logging of singles rates, HV, and ambient temperature for the full campaign. *Acceptance:* rejection of runs with rate drift > 10% within a run, HV excursion beyond the manufacturer's stability band, or temperature excursion > 5 °C; the rejection log is part of the published dataset.

4. Selection (locked)

- Trigger: three-fold coincidence — top paddle

× stopper × no bottom-paddle hit within a 200 ns veto window.

- **Rate guard:** runs with observed stopped rate > 10 min^{-1} are truncated at the rate-crossing point (keeps the dead-time correction within the verified ≤ 0.83% band).
- **Dead-time correction:** $N_{corr} = N_{obs}(1 + \lambda\tau_{dt})$; efficiencies with Clopper–Pearson intervals (rule of three for zero-failure sub-runs); rates with Garwood (exact Poisson) intervals [3].

5. Fit model and estimator (locked)

Analysis window $[t_{min}, t_{max}] = [0.5\ \mu s, 15.0\ \mu s]$. Per-event density:

$$f(t) = (1 - \varepsilon_b) \cdot e^{(-t/\tau)} / Z(\tau) + \varepsilon_b / (t_{max} -$$

t_min), where Z(τ) = τ (e^(−t_min/τ) − e^(−t_max/τ)).

- Unbinned extended maximum likelihood; ε_b profiled; parameters fitted in transformed coordinates (ln τ, logit ε_b).
- 68% interval from the profile likelihood (ΔlnL = ½, Wilks); Fisher interval as cross-check.
- **Late-pulse decision:** a second exponential term enters f(t) only if CD-3 shows ΔlnL > 5.9 for its inclusion (99% CL, 1 d.o.f.); otherwise the flat-background model stands.

6. Poisson certification (locked)

- **Fano statistic:** J = 40 equal-width bins; standardized residuals z_j = (n_j − μ_j)/√μ_j;

$F = \langle z^2 \rangle$ reported as $F \pm \sqrt{(2/J)}$ (Poisson expectation $F = 1$).

- **Goodness of fit:** parametric-bootstrap (Lilliefors-corrected) KS — $M = 10^4$ pseudo-spectra generated from the best-fit model, each refitted, p-value by exceedance of D. The plain KS p-value is not used because the model parameters are estimated from the data.
- **Acceptance criteria:** $|F - 1| \leq 3\sqrt{(2/J)}$ **and** $p \geq 0.05$. The same protocol is applied to the singles-rate log in fixed time windows to audit the counting channel during the run.

7. Unblinding gate (locked)

Before the measured $\hat{\tau}$ is examined, Monte-Carlo validation at $\tau = 2.1969811$ μs [1] with the

measured N and ε_b (10^4 pseudo-experiments; pipeline muon_lifetime_mc.py) must show: (1) |bias| < 0.2 σ_τ; (2) 68% profile-likelihood coverage within 3 binomial σ of nominal. Failure → protocol revision, addendum, analysis restart. Blinded: $\hat{\tau}$ and its interval. Open throughout: run logs.

8. Systematics plan (pre-specified)

- **Fit-window scan:** $t_{min} \in \{0.3, 0.5, 0.8, 1.2\}$ μs; $t_{max} \in \{10, 15, 20\}$ μs; maximum spread enters the systematic line.
- **Threshold scan:** as defined in CD-2.
- **Calibration audits:** as defined in CD-1, CD-4, CD-5.

9. Expected outcome (projection, not a claim)

With 10 min^{-1} stopped rate, 50% efficiency, and 30 days live time: $N \approx 2 \times 10^5$ decays, projected statistical precision $\sigma_\tau/\tau \approx 0.2\%$, total projected uncertainty $\approx 0.25\%$. The expected scientific output is: (i) a muon decay spectrum whose counts are certified Poisson by the Sec. 6 protocol; (ii) a lifetime estimate consistent with the PDG value [1], serving as an end-to-end validation of the certified counting chain; (iii) a fully documented calibration suite (CD-1 … CD-5) reusable by other small-scale muon instrumentation. These are projections and methodological deliverables, not precision claims on $\tau^+\mu$.

10. Amendments policy

Any change to selections, window, model, certification criteria, calibration protocols, or gates requires: (i) a versioned addendum (v1.2, v1.3, …) with a new time-stamp; (ii) restart of run numbering. Deviation without addendum invalidates the certification claims of the study.

11. Availability

Raw spectra, calibration datasets CD-1 … CD-5, run logs, analysis code (muon_lifetime_mc.py), and this pre-registration will be deposited on Zenodo under a new DOI at publication. Parent hardware verification: QCID-Sigma v5.4, Zenodo 10.5281/zenodo.23378709 [2].

References

[1] R. L. Workman et al. (Particle Data Group), PTEP 2022, 083C01 (2022); $\tau^+\mu$ = 2.1969811(22) μs.
[2] N. Khan, "QCID-Sigma v5.4 Final — Submission-Clean," Zenodo 10.5281/zenodo.23378709 (2026).
[3] F. Garwood, Biometrika 28, 437 (1936); C. J. Clopper and E. S. Pearson, Biometrika 26, 404 (1934).